<u>Descriptive statistics Assignment by PW skills</u>

Q.1. Understanding Central Tendency ?
A bakery tracks the daily sales of muffins (in dozens) over a week: [10, 12, 11, 15, 14, 13, 12].
What is the most representative value of their weekly sales, and why?

Ans : A good **most representative value** here is **12 dozen muffins**.

Why?

- When the data are ordered ([10, 11, 12, 12, 13, 14, 15]), the **median** (middle value) is **12**.
- **12** is also the **mode**, since it appears most often.
- The values are fairly balanced with no extreme outliers, so 12 reflects a *typical* day's sales well.

Q2: Mean in Real Life
A teacher records the marks of her students in a short quiz: [12, 15, 14, 16, 18, 20, 19].
What is the mean score, and what does it tell us about the class's performance?

Ans : Mean Score

The marks are: ([12, 15, 14, 16, 18, 20, 19])

- **Sum** = (12 + 15 + 14 + 16 + 18 + 20 + 19 = 114)
- **Number of students** = 7
- **Mean =** (114 ÷ 7 ≈ 16.3)

Note : The average score is about **16.3**, which suggests the class performed **quite well overall**. Most students scored in the mid-to-high teens, indicating generally strong understanding of the quiz material.

Q3: Mode in Real Life (Easy)
A store records the shoe sizes sold in one day: [7, 8, 9, 8, 8, 10, 7, 9].
What is the mode, and why is this information useful for the store manager?

Ans : Shoe sizes sold: ([7, 8, 9, 8, 8, 10, 7, 9])

- **Mode** = **8** (it appears most often)

Why this is useful:
Knowing that **size 8** sells the most helps the store manager:

- Stock more of that size
- Reduce lost sales due to running out
- Plan inventory more efficiently

In real life, the **mode** is especially useful for businesses because it shows the **most common customer preference**.

Q4: Median in Real Life
A car dealer notes the prices of used cars: [$8,000, $9,500, $10,200, $11,000, $50,000].
Why is the median a better measure than the mean in this case?
Calculate the median.

Ans : Car prices: ([$8,000, $9,500, $10,200, $11,000, $50,000])

- The data is already in order.
- The **median** (middle value) is **$10,200**.

Why the median is better than the mean:
The **$50,000** car is an **outlier** compared to the other prices. It would pull the **mean** upward and give a misleading impression of what a “typical” used car costs.
The **median** is not affected by extreme values, so **$10,200** better represents the usual price of cars at the dealership.

Q5: Dispersion Introduction

Puzzle times: ([25, 30, 27, 35, 40])

- **Maximum time** = 40 minutes
- **Minimum time** = 25 minutes
- **Range** = (40 − 25 = 15) minutes

What the range tells us:
The range shows that the student's puzzle-solving time varies by **15 minutes** from the fastest to the slowest day. This indicates a **moderate amount of variation**—some days the student finishes quickly, while on other days it takes significantly longer.

Q6: Range in Action
A farmer records the weekly weight of harvested apples (kg): [100, 105, 98, 110, 120].
Find the range. How can this help the farmer in planning his packaging?

Ans : Range in Action

Apple weights (kg): ([100, 105, 98, 110, 120])

- **Maximum** = 120 kg
- **Minimum** = 98 kg
- **Range** = (120 − 98 = 22) kg

How this helps the farmer:
The range shows that weekly harvests can vary by **22 kg**. Knowing this helps the farmer:

- Plan flexible **packaging sizes**
- Prepare extra boxes for high-yield weeks
- Avoid waste or shortages during lower-yield weeks

Q7: Variance for Decision-Making
Two delivery companies track delivery delays (in minutes).
Company A: variance = 6
Company B: variance = 15
Which company is more consistent, and why?

Ans : Variance for Decision-Making

- Company A: variance = **6**
- Company B: variance = **15**

More consistent company: Company A

Why:
Variance measures how spread out the delays are. A **smaller variance** means delivery times are **more consistent and predictable**. Since Company A has a lower variance, its delivery delays are more reliable than Company B's.

Q8: Standard Deviation in Context
A finance student compares the daily price fluctuations of two cryptocurrencies.
Coin A: standard deviation = $30
Coin B: standard deviation = $120
Which coin is riskier to invest in, and why?

Ans : Standard Deviation in Context

- Coin A: standard deviation = **$30**
- Coin B: standard deviation = **$120**

Riskier investment: Coin B

Why:
Standard deviation measures how much prices fluctuate. A **larger standard deviation means greater volatility**. Since Coin B's price

changes are much larger and less predictable, it carries **higher risk** than Coin A.

Q9: Combining Measures
A family records their monthly electricity usage (in kWh): [400, 420, 390, 450, 410].
Find the mean and standard deviation. What do these values together tell you about the family's energy use pattern?

Ans : Combining Measures (Hard)

Electricity usage (kWh): ([400, 420, 390, 450, 410])

- **Mean** = ((400 + 420 + 390 + 450 + 410) ÷ 5 = 414) kWh
- **Standard deviation ≈ 20.6 kWh**

What this tells us together:

- The family typically uses about **414 kWh per month**.
- The relatively **small standard deviation** shows their usage does not vary much from month to month.

➡️ This suggests a **fairly stable and consistent energy-use pattern**.

Q10: Practical Application
A basketball player's points in 8 games are recorded: [15, 18, 20, 22, 25, 17, 19, 21].
Find the mean, median, mode, range, and standard deviation. What insights can these measures provide
about the player's scoring performance?

Ans: Practical Application

Points scored: ([15, 18, 20, 22, 25, 17, 19, 21])
Sorted: ([15, 17, 18, 19, 20, 21, 22, 25])

- **Mean ≈ 19.6 points**
- **Median** = ((19 + 20) ÷ 2 = 19.5)
- **Mode** = **None** (all values are different)
- **Range** = (25 − 15 = 10)
- **Standard deviation ≈ 2.9 points**

Insights about performance:

- The mean and median being close shows the player scores **consistently around 19–20 points** per game.
- No mode suggests **no single repeated score**, indicating variety but not randomness.
- The small range and low standard deviation show **steady performance with little fluctuation**.

Note : Overall, the player is **reliable and consistent**, not overly dependent on unusually high or low scoring games.